

From PINNs to a quantity of interest

Goal-Oriented Deep Fourier Residual methods

Carlos Vera-Ciro

Goal-Oriented DFR series | Part 1

About this series

Four short decks, building the idea and then testing it honestly.

1. **The problem** (this deck): why the residual norm matters, and why global error is often the wrong target.
2. **The method**: goal-oriented DFR, a primal and an adjoint network trained together.
3. **The theory**: the error representation and the bound, with proofs.
4. **Results and limits**: what the method buys at equal compute, and where it does not help.

Takeaway: Audience: anyone comfortable with PDEs and neural networks. No prior DFR needed.

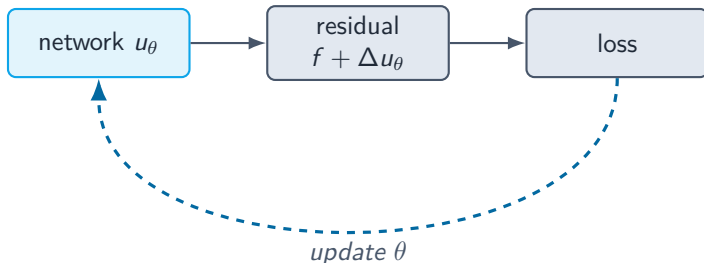
Section 1

Solving PDEs with networks

Solving a PDE with a neural network

We want u solving, say, Poisson's equation $-\Delta u = f$ with $u = 0$ on the boundary.

- Represent the unknown by a network u_θ with parameters θ .
- Choose θ so that u_θ satisfies the equation as closely as possible.
- “As closely as possible” is fixed by a **loss function**, and the choice of loss decides how faithful the answer is.



The trap: a small residual is not a small error

Physics-informed networks (PINNs) minimize the residual pointwise:

$$\frac{1}{N} \sum_i |f(x_i) + \Delta u_\theta(x_i)|^2.$$

- The **size** of the residual is not proportional to the **size** of the error.
- A residual small in L^2 can still hide a solution off in the places that matter, because differentiation amplifies high-frequency wiggles.

Takeaway: To control the error we must measure the residual in the right norm, not just make it pointwise small.

Section 2

The Deep Fourier Residual method

The DFR fix: the dual norm of the weak residual

Taylor, Pardo and Muga (2023): measure the weak residual in the H^{-1} **dual norm**.

- On a box, the sine functions diagonalize the Laplacian, so the dual norm is a weighted sum of the residual's sine coefficients,

$$\|R(u_\theta)\|_{V^*}^2 = \sum_k \frac{|\hat{R}_k|^2}{\lambda_k}, \quad \lambda_k = \sum_i (\pi k_i / L_i)^2,$$

computed fast by the Discrete Sine Transform.

- In $d \geq 2$ the eigenvalue couples the axes through the *sum* of squared frequencies, not a product of one-dimensional weights.

Error-loss equivalence (Taylor, Pardo, Muga 2023)

For a well-posed problem there are constants $\gamma, M > 0$ with

$$\frac{1}{M} \|R(w)\|_{V^*} \leq \|u^* - w\|_U \leq \frac{1}{\gamma} \|R(w)\|_{V^*}.$$

- **Two-sided:** the loss bounds the error above *and* below.
- Drive the DFR loss down, and the energy-norm error goes down with it.
- We prove this, and everything that follows, in Part 3.

Section 3

Quantities of interest

Global error is often not what we want

In practice the object of interest is a **single functional** of the solution, a quantity of interest (QoI):

a point value
 $u(x_0)$

a subdomain
average

a boundary
flux

- A small global error is *sufficient* but not *necessary* for an accurate QoI.
- Driving the global error to a tight tolerance is wasteful when only $J(u)$ matters.

Takeaway: We want a loss aimed at the quantity of interest, not the whole field. That is Part 2.